Quantum computing w/ neutral atoms

Why neutral atom qubits? (often Rb or Cs atom)

① Well defined qubit given by nature

^{87}Rb —— $F=2$ $\quad$ ^{133}Cs —— $F=4$

$I=3/2$ —— $F=1$ $\quad$ $I=7/2$ —— $F=3$

Hyperfine structure: $F = I + L + S$ ← electron spin
↑ ↑ → electron orbital
nuclear spin

^{87}Rb, ^{133}Cs → alkali atoms, single unpaired electron
ground states → S-states

$F = I + \cancel{L}^{0\ (S\text{-state})} + S$
↑ $= +/- 1/2$;

② Long lived, well-understood ground states for qubit.

- Recall Zeeman effect → $m_f = -F, -F+1 \ldots F$
  ↳ use $m_f = 0$ to avoid magnetic noise inducing decoherence (1st order cancellation)

③ Qubit lifetime typically seconds (limited by trap lifetime)

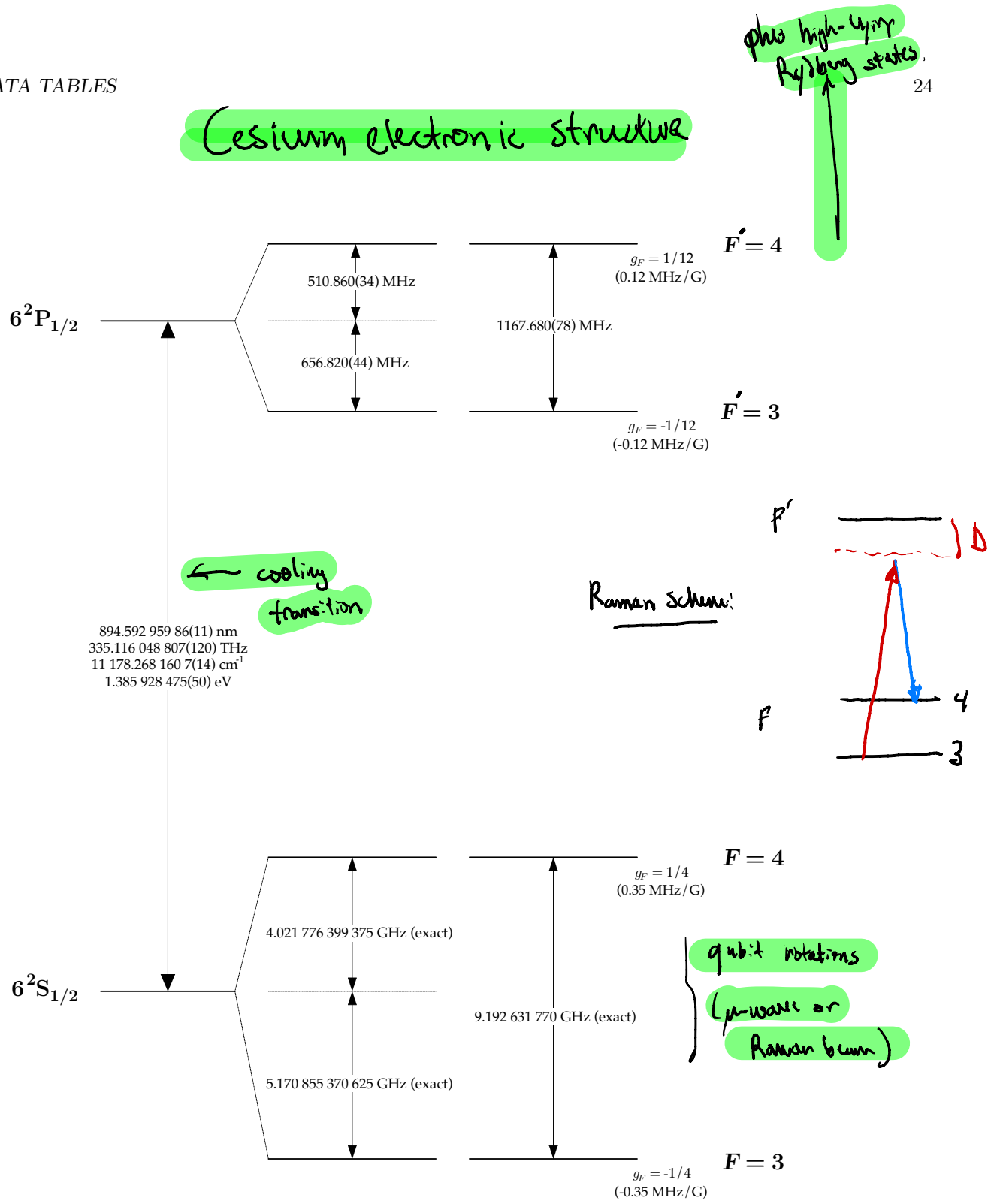


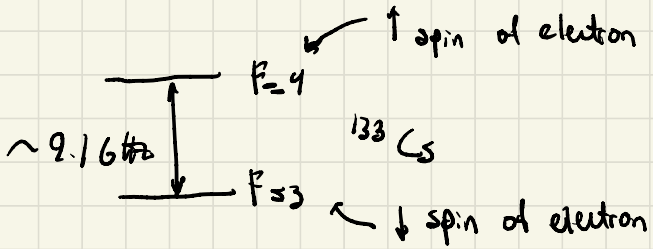
Figure 3: Cesium D_1 transition hyperfine structure, with frequency splittings between the hyperfine energy levels. The excited-state values are taken from [8, 22], and the ground-state values are exact, as a result of the current definition of the second. The approximate Landé g_F-factors for each level are also given, with the corresponding Zeeman splittings between adjacent magnetic sublevels.

④ Atoms provide homogeneous qubits:

- identical → simplifies control, but perhaps at the expense of cross talk between qubit operations.
- sensitive → potentially a disadvantage – AC Stark shift from external fields.

Qubit encoding:

(Typically) use "hyperfine qubit", encode in hyperfine ground states



$|0\rangle$ state → $|F=3\rangle$

$|1\rangle$ state → $|F=4\rangle$

Typical experimental values: (Bluvstein et al. Nature 604, 451-456 2022)

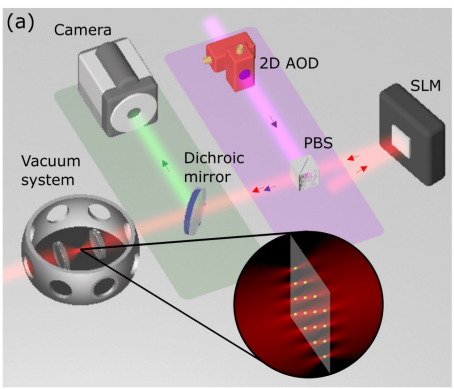
a) atom lifetime ~ 17s (background limited)

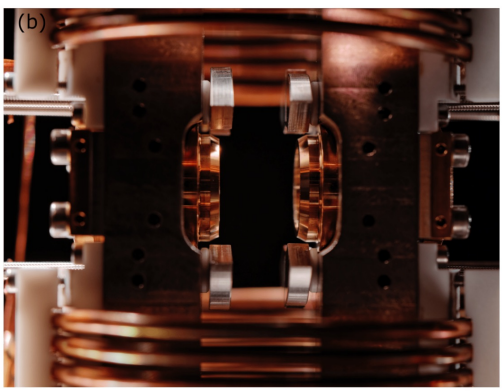
b) T_1 ~ 4 s (Background transitions)

c) $T_2^* = 4\,ms$ (finite temperature, other inhomogeneous broadening)

d) $T_2 = 1.5\,s$ enhanced by dynamical decoupling ($\sim$ spin echo)

Controllable assembly of many qubit registers





Elements

① Container for atoms → vacuum system $\leq 10^{-10}$ mBar or lower.

- either cryogenic or room temperature.

② Lasers for control of the system

a) laser cooling

$\vec{p} = m\vec{v}$

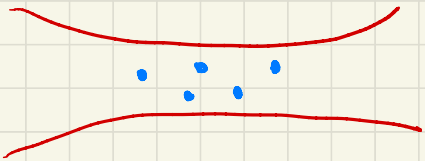
$\vec{p} = \hbar k$; $\nu \leq \omega_0 - \Delta$ (detuning)

- In Rb frame light shifts into resonance.

$\nu_{atom} = \left(1 + \frac{V}{c}\right)\nu_{photon}$

- Re-emission in random direction, leading to cooling.

b) lasers for trapping (dipole force)



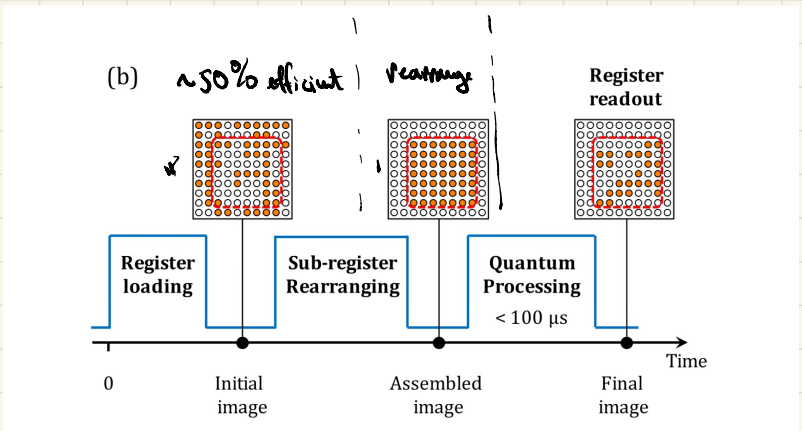
$$V(\vec{r}) = \frac{3\pi c^2}{2\omega_0^3}\frac{\Gamma}{\Delta} I(\vec{r})$$

(Γ: scattering rate $\sim \frac{1}{10\,\text{nsec}}$; $I(\vec{r})$: intensity pattern; Δ: detuning; ω_0: resonant frequency)

$$\Gamma_D(\vec{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\vec{r})$$

- ratio of terms goes like $1/\Delta$, so choose large detunings
- typically limits trap lifetime, also heating.

③ Assembling a multi-qubit register



④ Qubit operations (remainder of this lecture and Wed. lecture)

⑤ Readout

↳ use cooling light, only scatters if in $|1\rangle$ state.

Emerging Questions to consider

① How can we realize localized qubit operations?

② How can general circuits be realized?

How do we actually implement gates?

Single qubit operations

Review: Atom interacting w/ a classical monochromatic field.

Interaction Hamiltonian:

$H = H_0 + H_I$ ← interaction w/ field

↑ unperturbed Hamiltonian

Consider two-level system (i.e. ground states)

$$H_0 \psi_1(r) = E_1 \psi_1(r)$$
$$H_0 \psi_2(r) = E_2 \psi_2(r)$$

(stationary states)

$$\longrightarrow \psi_n(r,t) = \psi_n(r) e^{-iE_n t/\hbar} \; ; \; \frac{E_n}{\hbar} = \omega_n$$

Interaction Hamiltonian

$$H_I(t) = e \vec{r} \cdot \vec{E} \quad (\vec{r} \text{ is electron position})$$

$$\vec{E} = \frac{\vec{E}_0}{2} e^{i\omega t} + c.c.$$

$\omega \rightarrow$ exciting field frequency

- Interaction leads to mixing of two states.

In general

$$|\psi\rangle = \sum_n c_n e^{-i\omega_n t} |n\rangle$$

Plug in to $-i\hbar \frac{\partial|\psi\rangle}{\partial t} = H|\psi\rangle \Rightarrow$

$$\sum (\hbar\omega_n c_n + i\hbar \dot{c}_n) e^{-i\omega_n t}|n\rangle = \sum_n (\hbar\omega_n + H_I) c_n e^{-i\omega_n t}|n\rangle$$

Cancel common $\hbar\omega_n c_n$ terms. Multiply through by $\langle n|e^{i\omega_n t}$ and use orthonormality, change summation to over m

$$\hookrightarrow \quad i\dot{c}_n(t) = \frac{1}{\hbar}\sum_m \langle m|H_I|n\rangle e^{i\omega_{mn}t} c_m(t) \; ; \quad \omega_{mn} = \omega_m - \omega_n$$

So H_I couples states $|m\rangle$, $|n\rangle$ to each other, assuming $\langle m|H_I|n\rangle$ is non-zero.

For the two level system, get two coupled eqns:

$$i\dot{c}_1 = \Omega \cos(\omega t) c_2 e^{-i\omega_0 t} \quad (1)$$

$$i\dot{c}_2 = \Omega^* \cos(\omega t) c_1 e^{i\omega_0 t} \quad (2)$$

Definitions:

$\omega_0 = \omega_1 - \omega_2$ (transition frequency)

$\Omega = \langle 1|e\vec{r}\cdot\vec{E}_0|2\rangle/\hbar$ (Rabi frequency)

$\Delta = \omega - \omega_0$

Expand cosine to give

$$i\dot{c}_1 = \frac{\Omega c_2}{2}\left[e^{i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}\right]$$

Rotating wave approximation:

drop term w/ $\omega + \omega_0$, keep slowly varying term.

Define $c_2(t) = -\tilde{c}_2(t)\, e^{i\Delta t}$

Rewriting:

$$i\dot{c}_1 = \frac{\Omega}{2}\tilde{c}_2 \qquad (1b)$$

$$i\dot{\tilde{c}}_2 = -\frac{\Omega^*}{2}c_1 + \Delta\tilde{c}_2 \qquad (2b)$$

Trial solution: $c_1(t) = e^{-iKt} \longrightarrow \tilde{c}_2(t) = \frac{2K}{\Omega}e^{iKt}$

(plug in to 1b)

Plug in to 2b:

$$2(i\cdot -i)\frac{K^2}{\Omega}e^{iKt} = \frac{\Omega^*}{2}e^{-iKt} - \frac{2\Delta K}{\Omega}e^{-iKt}$$

$$\longrightarrow K^2 + \Delta K - \left|\frac{\Omega}{2}\right|^2 = 0.$$

$$K_\pm = \frac{1}{2}\left(-\Delta \pm \sqrt{\Delta^2+\Omega^2}\right)$$

$$= \frac{1}{2}\left(-\Delta \pm \Omega'\right)$$

$\Omega' = \sqrt{\Delta^2+\Omega^2}$

Generalised Rabi frequency

General solutions for c_1 and $\tilde{c}_2$:

$$c_1(t) = Ae^{iK_+t} + Be^{iK_-t}$$

$$\tilde{c}_2(t) = -\frac{2}{\Omega}\left(AK_+e^{iK_+t} + BK_-e^{iK_-t}\right)$$

Now plug back in K to determine $c_1(t)$, $c_2(t)$

Initial conditions: $c_1(0) = 1$; $c_2(0) = 0$

$$c_1(t) = e^{i\Delta t/2}\left[\cos(n't/2) + \frac{i\Delta}{n'}\sin(n't/2)\right]$$

$$c_2(t) = -\frac{in}{n'}e^{i\Delta t/2}\sin(n't/2)$$

Probability of finding atom in the second state:

$$|c_2(t)|^2 = \left(\frac{n}{n'}\right)^2 \sin^2(n't/2)$$

Rabi oscillations.

↳ for $\omega = \omega_0$, $\Delta = 0$ and get

$$|c_2(t)|^2 = \sin^2\left(\frac{nt}{2}\right)$$

More generally, one can write the evolution as a matrix transforming

$$\vec{c}(t_0) = \begin{pmatrix} c_1(t_0) \\ c_2(t_0) \end{pmatrix}$$

$$\vec{c}(t) = U(t-t_0)\,\vec{c}(t_0)$$

For $t_0 = 0$ $U(t) =$

$$\begin{pmatrix} e^{i\Delta t/2}\left[\cos\left(\frac{\Omega' t}{2}\right) + \frac{i\Delta}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right)\right] & -ie^{i\Delta t/2}\frac{\Omega^*}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right) \\ -ie^{-i\Delta t/2}\frac{\Omega}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right) & e^{-i\Delta t/2}\left[\cos\left(\frac{\Omega' t}{2}\right) - \frac{i\Delta}{\Omega'}\sin\left(\frac{\Omega' t}{2}\right)\right] \end{pmatrix}$$

Note: Recall $e^{i\Delta t/2} = e^{i(\omega-\omega_0)t/2}$. We could apply an arbitrary phase φ_0 to the driving field. This results in a value of $e^{i\varphi_0}$ being applied to the terms in the matrix.

For (ideal) qubit operation, we consider $\Delta = 0$

(note this will be often affected by noise or other aspects!

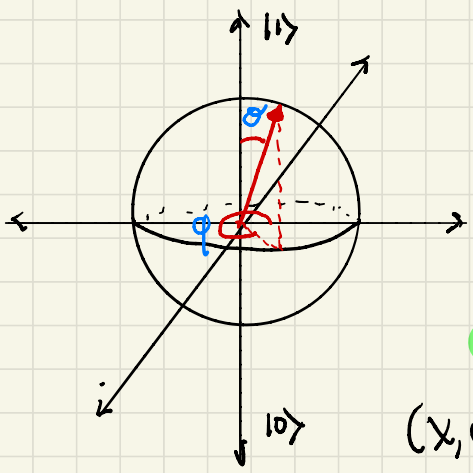
Write $\vec{C}(t) = \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix}$

Then $C(t) = U(t-t_0)\,C(t_0)$; where

$$U(t) = \begin{pmatrix} \cos\left(\frac{\Omega t}{2}\right) & -i\sin\left(\frac{\Omega t}{2}\right) \\ -i\sin\left(\frac{\Omega t}{2}\right) & \cos\left(\frac{\Omega t}{2}\right) \end{pmatrix}$$

- So now we only need to define the right $\frac{\pi}{2}$ to realise a variety of transformations

Bloch sphere picture again: (quantisation defined by IJG field)



State vector:

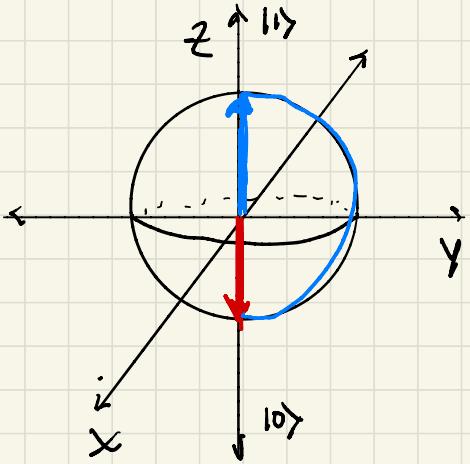
$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$

Cartesian Version:

$$(x, y, z) = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$$

Key transformations:

① π pulse



Apply μ-wave pulse for

$$t = \frac{\pi}{|\Omega|}$$

$$U_\pi = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$$

X-gate. (with global phase of $e^{-i\pi/2}$)

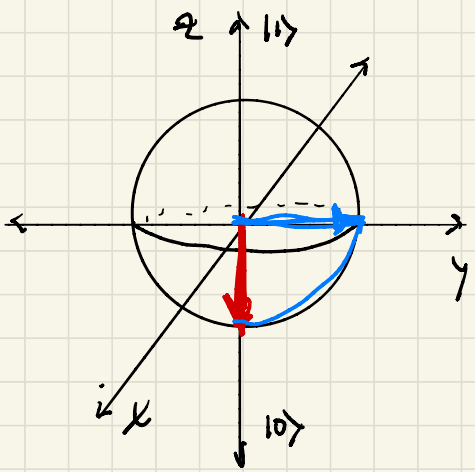
② 2π pulse:

$$U_{2\pi} = U_\pi U_\pi = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

↑ note phase shift.

(3) $\pi/2$ pulse.

Apply pulse for time $t = \frac{\pi}{2\Omega}$:



$$U_{\pi/2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$$

How can one realize other gates? I.E. Y gate?

- Change phase of driving field: a phase shift of $\pi/2$ will give this:

$$U(t) = \begin{pmatrix} 0 & -e^{-i\pi/2} i \\ -e^{i\pi/2} i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

(on resonance) set $t = \frac{\pi}{\Omega}$

This is a Y-gate w/ the same global phase shift:

$$e^{+i\pi/2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \longrightarrow \text{Y-gate}$$

What about z rotations?

up to an arbitrary phase α, these can be decomposed into combinations of the other axes:

$$R_z(\theta) = e^{i\alpha} R_x(\beta) R_y(\gamma) R_x(\delta)$$

For example:

$$R_z(\pi) = R_x(3\pi/2) R_y(\pi) R_x(\pi/2)$$

$$R_z(\pi/2) = R_x(\pi/2) R_y(3\pi/2) R_x(3\pi/2)$$

This then means we can get arbitrary rotations on the Bloch sphere → easily implement any 1-qubit gate!

- Alternatively one can introduce a known differential Stark shift between the energy levels which is effectively:

$$U(\delta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix}$$

index	x axis	y axis	z axis	U	R_3	R_2	R_1	θ_{total}
1	I	I	I	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	-	-	-	0
2	I	I	$\pi/2$	$e^{-\imath\pi/4}\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$R_x(\pi/2)$	$R_y(\pi/2)$	$R_x(3\pi/2)$	$5\pi/2$
3	I	I	π	$-i\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	-	$R_x(\pi)$	$R_y(\pi)$	2π
4	I	I	$-\pi/2$	$e^{\imath\pi/4}\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$	$R_x(\pi/2)$	$R_y(3\pi/2)$	$R_x(3\pi/2)$	$7\pi/2$
5	I	π	I	$-1\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	-	-	$R_y(\pi)$	π
6	I	π	$\pi/2$	$-e^{\imath\pi/4}\begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix}$	$R_x(\pi/2)$	$R_y(\pi/2)$	$R_x(\pi/2)$	$3\pi/2$
7	π	I	I	$-i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	-	-	$R_x(\pi)$	π
8	π	I	$\pi/2$	$e^{-\imath\pi/4}\begin{pmatrix} 0 & 1 \\ -i & 0 \end{pmatrix}$	$R_x(\pi/2)$	$R_y(3\pi/2)$	$R_x(\pi/2)$	$5\pi/2$
9	π	$\pi/2$	I	$\frac{-i}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	-	$R_x(\pi)$	$R_y(\pi/2)$	$3\pi/2$
10	I	$-\pi/2$	I	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$	-	-	$R_y(3\pi/2)$	$3\pi/2$
11	$\pi/2$	I	$\pi/2$	$\frac{e^{-\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$	-	$R_y(3\pi/2)$	$R_x(\pi/2)$	2π
12	$\pi/2$	π	$\pi/2$	$-\frac{e^{\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$	-	$R_y(3\pi/2)$	$R_x(3\pi/2)$	3π
13	π	$-\pi/2$	I	$\frac{i}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$	-	$R_x(\pi)$	$R_y(3\pi/2)$	$5\pi/2$
14	$-\pi/2$	I	$\pi/2$	$\frac{e^{-\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ i & i \end{pmatrix}$	-	$R_y(\pi/2)$	$R_x(3\pi/2)$	2π
15	I	$\pi/2$	I	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$	-	-	$R_y(\pi/2)$	$\pi/2$
16	$-\pi/2$	π	$\pi/2$	$\frac{e^{\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ -i & -i \end{pmatrix}$	-	$R_y(\pi/2)$	$R_x(\pi/2)$	π
17	$-\pi/2$	$-\pi/2$	I	$\frac{e^{-\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & i \\ -1 & i \end{pmatrix}$	-	$R_x(3\pi/2)$	$R_y(3\pi/2)$	3π
18	$-\pi/2$	$\pi/2$	I	$\frac{e^{\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$	-	$R_x(3\pi/2)$	$R_y(\pi/2)$	2π
19	$-\pi/2$	π	I	$\frac{i}{\sqrt{2}}\begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix}$	-	$R_x(3\pi/2)$	$R_y(\pi)$	$5\pi/2$
20	$-\pi/2$	I	I	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$	-	-	$R_x(3\pi/2)$	$3\pi/2$
21	$\pi/2$	$-\pi/2$	I	$\frac{e^{\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ -1 & -i \end{pmatrix}$	-	$R_x(\pi/2)$	$R_y(3\pi/2)$	2π
22	$\pi/2$	I	I	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$	-	-	$R_x(\pi/2)$	$\pi/2$
23	$\pi/2$	π	I	$\frac{-i}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ i & -1 \end{pmatrix}$	-	$R_x(\pi/2)$	$R_y(\pi)$	$3\pi/2$
24	$\pi/2$	$\pi/2$	I	$\frac{e^{-\imath\pi/4}}{\sqrt{2}}\begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}$	-	$R_x(\pi/2)$	$R_y(\pi/2)$	π

TABLE I. Elements of the Clifford group $\mathcal{C}_1$ for a single qubit. The notation θ in column 2,3 or 4 is shorthand for $R_j(\theta)$ about the corresponding axis. The resulting operator U is due to the sequence $U_xU_yU_z$, reading right to left. The operators have been factored so that the first nonzero element in the top row is unity. The microwave implementation $R_3R_2R_1$, reading right to left, used the rotations shown in the last three columns. These give the same Clifford operations as U, up to irrelevant global phase factors.

Single qubit gate fidelities:

upto "4 nines" 0.9999(3)

from. T. Xia et. al. (P.R.L. 114, 100503 2015)

What fidelities are relevant?

Relevant Work	Code	Threshold	Geometric Constraints	Architectural context.
(ABO08)	$[[7,1,3]]$	$O(10^{-6})$	A.I.[a]	None
(Got97)	$[[7,1,3]]$	$O(10^{-4}-10^{-6})$	A.I.	None
(KLZ96)	$[[7,1,3]]$	$O(10^{-6})$	A.I.	None.
(PR12)	$[[23,1,7]]$	$O(10^{-3})$	A.I.	None
(Kni05)	4- and 6-qubit detection	$O(10^{-2})$	A.I.	None.
(BAO+12)	Toric Code	18%	2D NN[b] array, P.B.C.[c]	None, Theoretical upper bound[d]
(MCT+04)	$[[7,1,3]]$[e]	$O(10^{-4})$	A.I. with movement penalty	Specific to Ion-Traps
(SFH08)	$[[7,1,3]]$	$O(10^{-6})$	Bilinear NN array	Kane P:Si system (Kan98)
(SBF+06)	$[[7,1,3]]$	$O(10^{-7})$	variable width NN array	General NN systems
(SDT07)	$[[7,1,3]]$	$O(10^{-5})$	2D NN arrays	General NN systems
(FTY+07)	$[[7,1,3]]$	$O(10^{-6})$	Bilinear NN array	Superconducting qubits
(BKSO05)	$[[7,1,3]]$	$O(10^{-9})$	A.I. with movement penalty	Ion-Traps
(RHG07)	Topological Cluster	$O(10^{-2}-10^{-3})$	3D NN array	Photonic Qubits (DFS+09)
(WFSH10)	Surface Codes	$O(10^{-2}-10^{-3})$	2D NN array	Quantum Dots, Diamond (JMF+12; YJG+12)

[a] Arbitrary interactions
[b] Nearest Neighbour
[c] Periodic Boundary conditions
[d] This result is calculated assuming perfect quantum gates and is known as the code capacity.
[e] Only a preliminary estimate

So the fidelities are comp. to Quant. Dots, photonic qubits.

(As of 2013 "Quantum error correction for beginners")

Single gate solution done with "Stark shift beam"

